

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Electrical contact cleaner, Part Number 3824510, or equivalent
- Fuel System Clean Care Kit, Part Number 4919073
- Fuel Pump Gear Replacer Kit, Part Number 4919586
- Gear puller, Part Number ST-647, or equivalent
- Fuel pump timing tool, Part Number 5572664

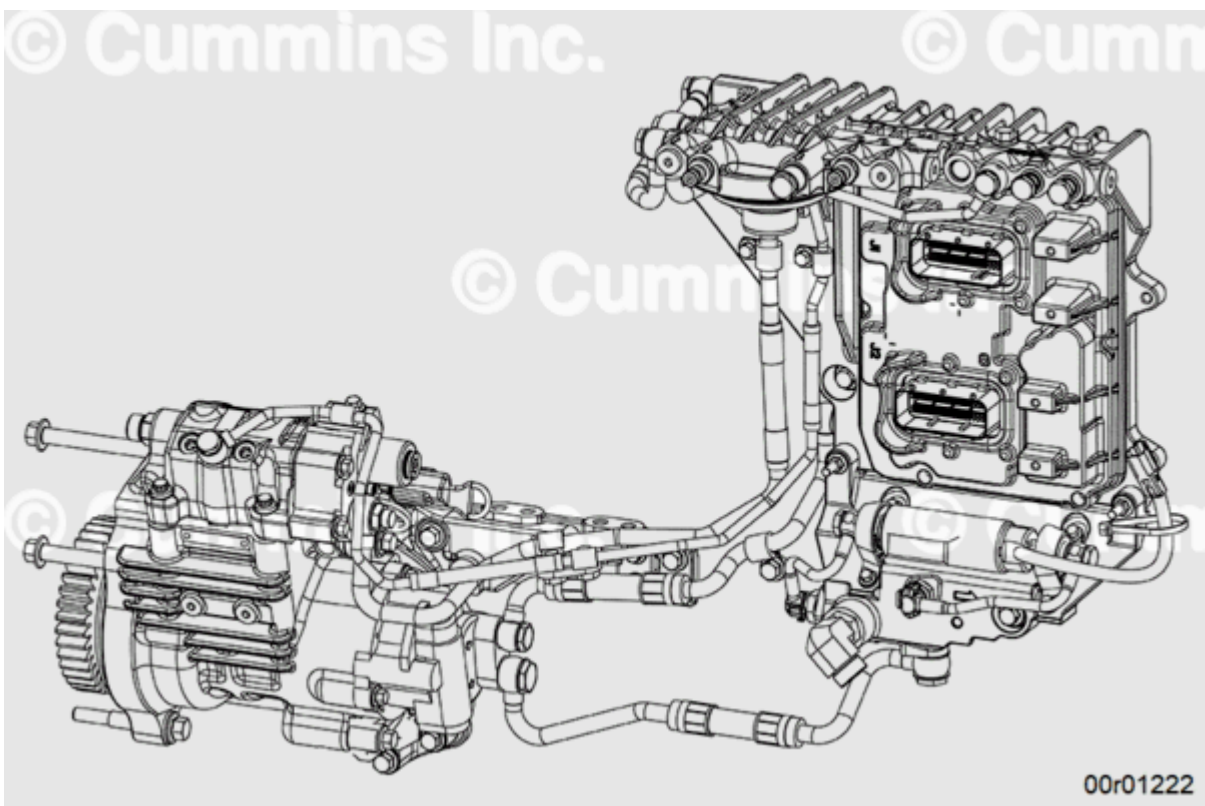
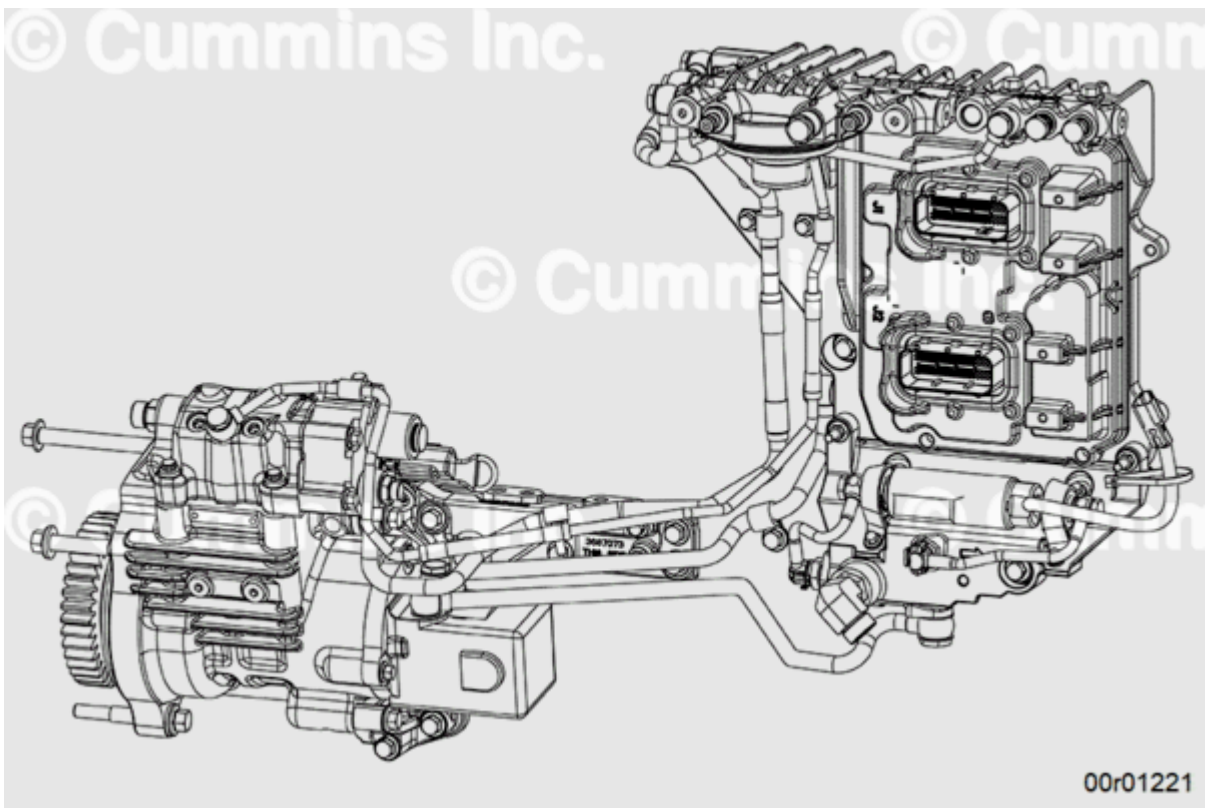
Additional Service Items

- No additional service items required.

General Information

Engines can be equipped with two fuel pump assembly variations. These fuel pumps will have the same performance despite gear pump visual variation(s). Each fuel pump requires distinct fuel line configurations.

Cummins® Service Fuel Pumps:



Preparatory Steps

⚠ WARNING ⚠

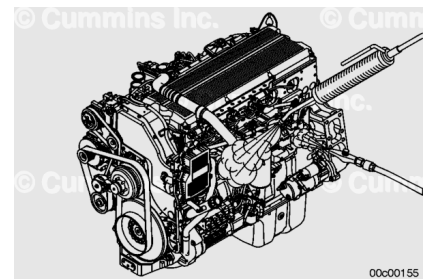
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.



Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.), which would expose the fuel system or internal engine component to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 in Section 0.

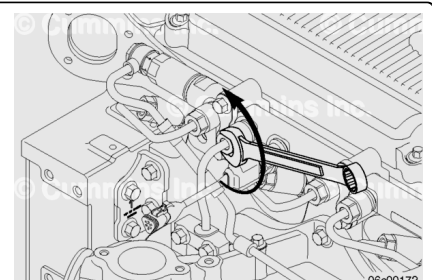
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Clean the fuel lines connected to the fuel pump, fuel pump gear pump, and mating components with electrical contact cleaner, Part Number 3824510, or equivalent.

To prevent damage from debris and contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

Normal engine operation creates highly pressurized fuel in the fuel line which will remain in the fuel line after engine shutdown. Never open the fuel system when the engine is operating. Before servicing the fuel system, always loosen



the pump-to-rail fuel line at the rail to vent the pressure. Keep hands clear of the line when loosening. High-pressure fuel spray can penetrate the skin, resulting in serious personal injury or death.

⚠ WARNING ⚠

When servicing the engine do not use the starting motor to rotate the engine with a high-pressure fuel system joint open. Rotating the engine can create highly pressurized fuel in the fuel system. High-pressure fuel spray can penetrate the skin, resulting in serious personal injury or death.

Note : A machined slot in this fitting directs the fuel spray toward the cylinder block.

Before servicing the fuel system, loosen the pump-to-rail line at the rail to vent the fuel pressure. Refer to Procedure 006-081 in Section 6. (</qs3/pubsys2/xml/en/procedures/337/337-006-081.html>)

Keep hands clear of the line when loosening the fuel rail nut.

Tighten the fuel rail nut.

Torque Value: 47 n•m [35 ft-lb]

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

- Disconnect the batteries. See equipment manufacturer service information.

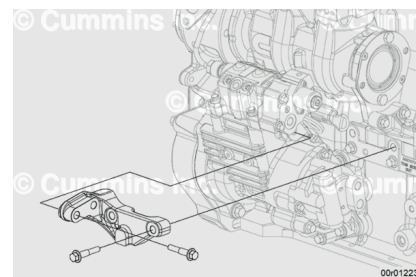
- Rotate the crankshaft **clockwise** until the "insert pin" mark on the outside diameter of the engine vibration damper is aligned with the mark on the gear housing. Refer to Procedure 000-017 in Section 0. (</qs3/pubsys2/xml/en/procedures/377/377-000-017.html>)
- Remove the fuel supply lines from the fuel pump gear pump. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-024.html>)
- Disconnect the engine harness from the fuel pump actuator.
- Remove the fuel supply line from the fuel pump. Mark the banjo bolt for reuse in this location. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-024.html>)
- Remove the fuel drain line from the fuel pump. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-013.html>)
- Disconnect the pump-to-rail line from the fuel pump. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-051.html>)

Note : The fuel line from the engine control module (ECM) cooling plate to the high-pressure fuel pump contains a filter screen at the high-pressure fuel pump banjo fitting. This screen should be examined and cleaned or the banjo bolt replaced if the screen is damaged.

Remove

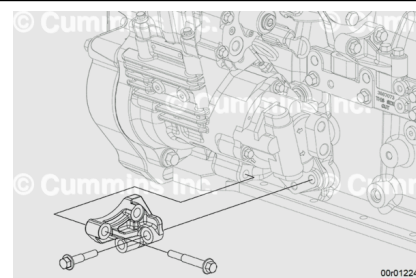
Remove the three capscrews from the fuel pump tail support bracket to the cylinder block and mounting bracket to fuel pump brace.

Remove the two capscrews from the fuel pump tail support bracket to fuel pump.



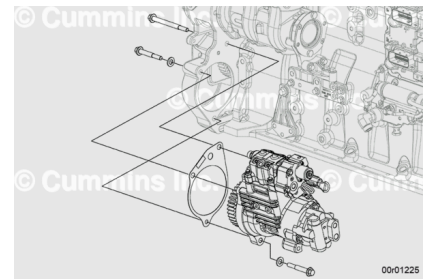
Remove the fuel pump mid-support bracket by removing the bolts connecting the mid-support bracket to the cylinder block and fuel pump brace.

Remove the capscrews connecting the mid-support bracket to the fuel pump.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the three mounting capscrews that hold the fuel pump to the gear housing.

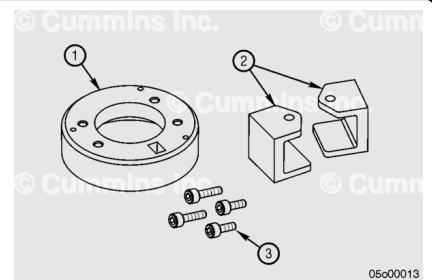
Remove the fuel pump. Discard the gasket.

Disassemble

Fuel Pump Gear Replacer Kit, Part Number 4919586, is required to remove and install the fuel pump gear.

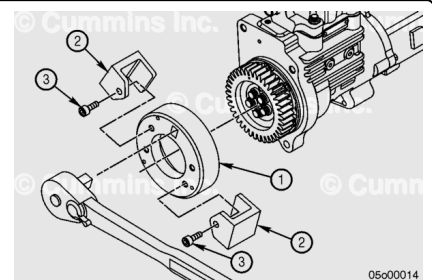
The kit consists of the following components:

1. Gear replacer
2. Replacer jaws
3. Socket head capscrews.



Mount the gear replacer (1), replacer jaws (2), capscrews (3), and a breaker bar to the fuel pump gear.

Install the capscrews hand-tight.

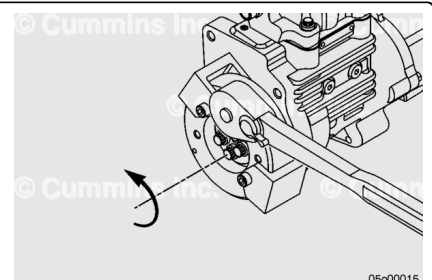


⚠ CAUTION ⚠

It will be necessary to re-time the fuel pump if the camshaft is rotated.

⚠ CAUTION ⚠

Excessive axial movement of the fuel pump camshaft may damage the fuel pump.

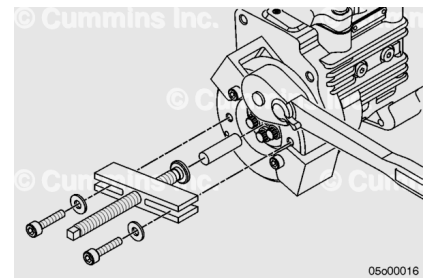


Hold the fuel pump gear stationary while loosening the five capscrews attaching the fuel pump gear to the fuel pump shaft. Do **not** remove the capscrews from the fuel pump shaft.

The capscrews are needed to prevent the gear and service tool from falling during fuel pump gear removal.

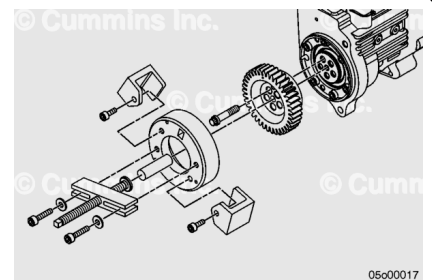
Attach gear puller, Part Number ST-647, or equivalent to the gear replacer. Use the two long capscrews included in the Fuel Pump Gear Replacer Kit, Part Number 4919586. It will be necessary to use a dowel inserted into the center of the gear to press against the fuel pump shaft.

Turn the forcing screw against the dowel and fuel pump shaft nose until the fuel pump gear detaches from the shaft.



Remove the gear replacer and the gear puller.

Remove the fuel pump gear retaining capscrews and the fuel pump gear.

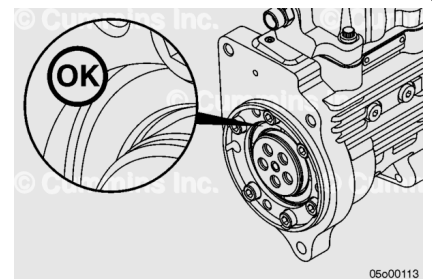


Inspect for Reuse

Inspect the fuel pump front bushing. A good bushing will be silver/grey in color with square edges. The camshaft **must** be centered in the bushing, with no gaps visible.

Replace the fuel pump if:

- The fuel pump housing around the bushing appears to be blued, burned, or shows signs of excessive heat.
- The fuel pump front bushing appears to be blued, burned, or shows signs of excessive heat.
- The fuel pump bushing is melted or appears to be extruded.
- Gaps can be seen between the bushing and the fuel pump housing, the bushing and camshaft, or both.

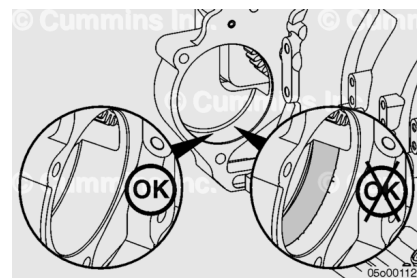


Inspect the fuel pump cavity in the front gear housing for damage.

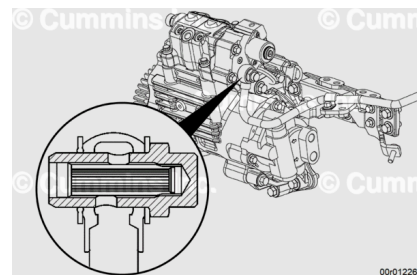
The bottom of the cavity **must** be smooth. If a step is machined in the bottom of the fuel pump cavity, it indicates the fuel pump gear has contacted the housing. The front gear

housing **must** be replaced and the fuel pump inspected for bushing damage, as outlined above.

The lubricating system **must** also be inspected for aluminum debris.



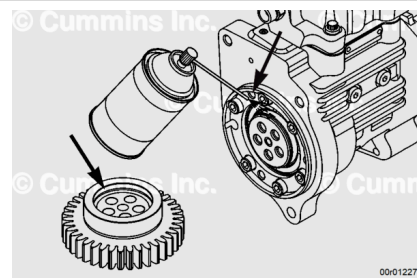
Inspect the banjo bolt filter for debris. This screen should be examined and cleaned, or the banjo bolt replaced, if the screen is damaged.



Assemble

Clean the oil drilling, the nose of the drive shaft, and the fuel pump gear inside diameter. Use electrical contact cleaner, Part Number 3824510, or equivalent.

The fuel pump drive gear inside diameter and the drive shaft outside diameter **must** be clean and dry before installing the gear.



Place the fuel pump drive gear over the fuel pump shaft.

Install the gear replacer (1), replacer jaws (2), and short capscrews (3) from Fuel Pump Gear Replacer Kit, Part 4918586, onto the fuel pump gear.

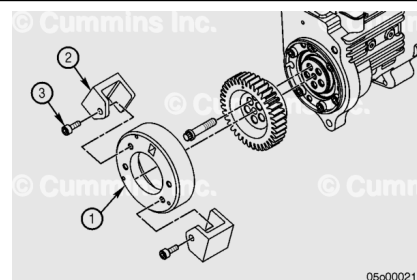
Attach a breaker bar to the gear replacer.

Tighten the capscrews in a crisscross pattern.

Torque Value:

1. 25 n•m [221 in-lb]
2. Rotate 60 degrees.

Remove the Fuel Pump Gear Replacer Kit components and the breaker bar.



Time

⚠ CAUTION ⚠

The timing pin must be installed before the fuel pump is removed and must remain in place until the fuel pump is fully installed and tightened. Failure to install the timing pin can lead to improper timing of the fuel pump, resulting in excessive noise.

⚠ CAUTION ⚠

The fuel pump timing gauge must be removed prior to engine operation. Failure to remove the fuel pump timing gauge can lead to severe fuel pump damage.

Note : The following is a representation of gear train impact noise producing a "freight train" or "helicopter" noise.

http://tsb.cumminsvirtualcollege.com/TSB170124_en.aspx
(http://tsb.cumminsvirtualcollege.com/TSB170124_en.aspx)

Pin the crankshaft. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)

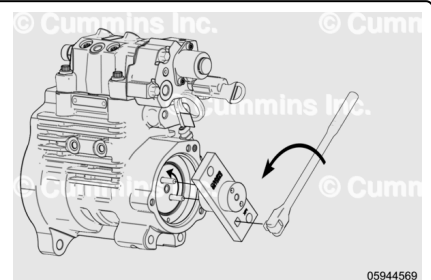
Remove the high-pressure fuel pump. See the Remove step in this procedure.

Remove the fuel pump gear pump from the rear of the high-pressure fuel pump. Refer to Procedure 005-025 in Section 5. (/qs3/pubsys2/xml/en/procedures/377/377-005-025.html)

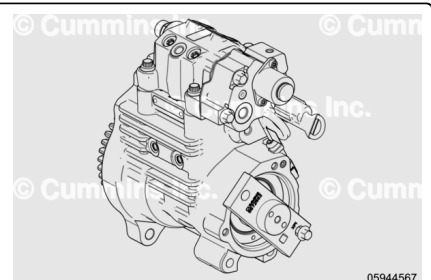
Insert fuel pump timing gauge, Part Number 5572562, onto the fuel pump housing. Align the slots on the fuel pump timing gauge with the fuel pump camshaft pins. Verify the XPI stamping is visible when the fuel pump timing gauge is installed.

Insert a 3/8 inch breaker bar into the barring adapter located on the front of the fuel pump timing gauge.

Rotate the fuel pump camshaft to align the gauge's XPI hole with the fuel pump bolt hole closest to the engine.

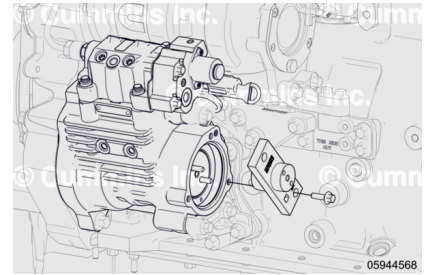


Insert the timing gauge capscrew into the aligned holes, turning it **clockwise** to engage a few threads. Do **not** tighten the bolts all the way. This will allow the timing gauge to float and the gear teeth to more easily mesh during fuel pump installation.

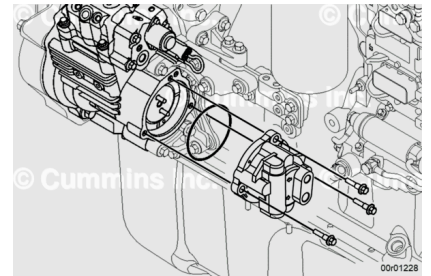


Mount the fuel pump to the engine. See the Install step in this procedure.

Remove the capscrew and fuel pump timing gauge from the fuel pump camshaft.



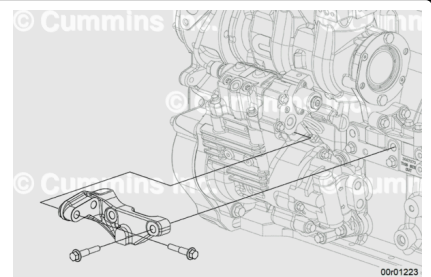
Install the fuel pump gear pump. Refer to Procedure 005-025 in Section 5. (/qs3/pubsys2/xml/en/procedures/377/377-005-025.html)



Install

⚠ WARNING ⚠

When servicing the engine do not use the starting motor to rotate the engine with a high-pressure fuel system joint open. Rotating the engine can create highly pressurized fuel in the fuel system. High-pressure fuel spray can penetrate the skin, resulting in serious personal injury or death.



The fuel pump **must** be timed before installation is completed. See the Time step in this procedure.

Insert the top two mounting capscrews through the gear housing and place the new fuel pump gasket over the capscrews.

Position the fuel pump over the mounting studs and engage the fuel pump gear.

Never rotate the fuel pump while meshing the fuel pump gear teeth with the fuel pump idler gear. Rotating the fuel pump while meshing gears will lead to mistiming the fuel pump.

Install the fuel pump mounting capscrews.

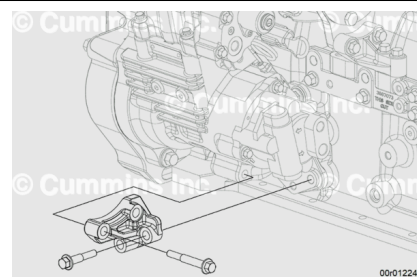
Torque Value: 81 n•m [60 ft-lb]

If necessary, install the fuel pump braces. Refer to Procedure 005-235 in Section 5.

(/qs3/pubsys2/xml/en/procedures/377/377-005-235.html)

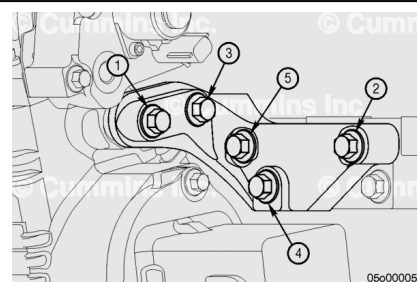
Install the fuel pump tail support bracket.

Install the mounting capscrews hand-tight.



Tighten the capscrews in the sequence shown.

Torque Value: 60 n•m [44 ft-lb]

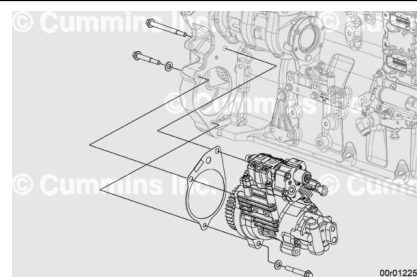


Install the fuel pump mid-support bracket.

Install the mounting capscrews hand-tight.

Tighten the capscrews.

Torque Value: 60 n•m [44 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Hold banjo fittings while tightening the banjo bolt to prevent fitting rotation. Allowing the banjo fitting to rotate may damage the fuel line.

- Install the fuel supply lines to the fuel pump. Make certain to install the marked banjo bolt back in its original position. Refer to Procedure 006-024 in Section 6.

(/qs3/pubsys2/xml/en/procedures/377/377-006-024.html)

- Install the fuel drain line to the fuel pump. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-013.html)
- Connect the wiring harness to the fuel pump actuator. Refer to Procedure 019-117 in Section 19. (/qs3/pubsys2/xml/en/procedures/377/377-019-117.html)
- Prime the fuel system. Refer to Procedure 005-234 in Section 5.
(/qs3/pubsys2/xml/en/procedures/377/377-005-234-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks and proper operation.